

SIN-topology rigidity and absence of locally compact images for projective unitary groups of factors

Candidate partial-result packet for arXiv:1804.09515

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Status. Candidate partial result; likely valid, subject to expert review. The theorem settles the source question for separable SIN topologies and gives a stronger algebraic obstruction to locally compact target groups. It does not settle uniqueness among arbitrary separable group topologies.

1 Source question and partial answer

Dowerk asks in Section 5, pages 11–12 of [1] whether $U(M)$ or $PU(M)$ has a unique nontrivial separable group topology when M is a separable full von Neumann algebra, and whether a coarsest separable group topology exists more generally. For first-countable topologies on $PU(M)$ he reformulates this as completeness of every bounded compatible left-invariant metric inducing a separable topology.

pointwise norm convergence on the predual M_* . As in the proof of [30, Theorem XIV.3.8] we have that $\text{PU}(M)$ is isomorphic to $\text{Inn}(M)$ under Ad . Since $\text{Inn}(M)$ is not closed in $\text{Aut}(M)$, it cannot be Polish by [3, Proposition 1.2.1]. Using the fact that a subspace of a separable metric space is separable, we obtain a separable group topology on $\text{PU}(M)$ which is not Polish. $\square$

Examples of full factors are the group factors $L(\mathbb{F}_n)$ of the free group $\mathbb{F}_n$ in $n \geq 2$ generators by [25, Chapter VI]. An example of a non-full II_1 factor is the hyperfinite II_1 factor $\mathcal{R}$ by [25, Chapter VI], and hence $\text{PU}(\mathcal{R})$ has at least two nontrivial separable group topologies - one being Polish (the strong operator topology induced by the 2-norm of the trace) and one only being separable.

QUESTION. Does $\text{U}(M)$ (or $\text{PU}(M)$) have a unique nontrivial separable group topology whenever M is a separable full von Neumann algebra? Does $\text{U}(M)$ (or $\text{PU}(M)$) have a coarsest separable group topology for any infinite-dimensional separable von Neumann algebra?

We remark that it is known from [10, Theorem 1.4] that the projective unitary group of any separable II_1 factor has a unique Polish group topology.

Recall that a topological group has *automatic continuity* if every homomorphism to any separable topological group is continuous. A positive answer to the first question would imply automatic continuity for the group in question, endowed with the strong operator topology. Similarly, a positive answer to the second question would imply automatic continuity of the given group with respect to its coarsest nontrivial separable topology. Note that it was proven in [10, Theorem 1.4] that whenever M is a separable finite factor (i.e. M is finite-dimensional or of type II_1), every homomorphism from $\text{PU}(M)$ to any separable SIN group is continuous.

Let us rephrase the question above in the case that M is either a separable II_1 factor or a separable type III factor. If G is a topologically simple group with neutral element 1_G , then the closure of $\{1\}$ is a closed normal subgroup of G , and hence any nontrivial group

nontrivial group topology $\mathcal{T}$ on $\text{PU}(M)$ is Hausdorff. Assume that $\mathcal{T}$ is first countable. By the Birkhoff-Kakutani Theorem [21, Theorem 9.1] the topology $\mathcal{T}$ is metrizable with a compatible left-invariant metric d . Hence d is bounded by Corollary 1.3. Thus we can reformulate our question as follows:

Is a bounded left-invariant metric d inducing a separable topology on $\text{PU}(M)$ automatically complete for $\text{PU}(M)$ whenever M is a separable full II_1 factor or a separable type III factor?

Acknowledgments

The result below concerns $\text{PU}(M) = \text{U}(M)/\mathbb{T}$. Recall that a topological group is *SIN* if it has a neighbourhood basis at the identity consisting of conjugacy-invariant sets, and is *precompact* if its completion is compact.

Theorem 1 (SIN rigidity). *Let M be a factor with separable predual and put $G = \text{PU}(M)$.*

1. *If M is of type II_1 , the strong operator topology is the unique nontrivial separable SIN group topology on the abstract group G .*
2. *If M is of type III, the abstract group G admits no nontrivial separable SIN group topology.*

3. In either case, G admits no Hausdorff precompact group topology. Equivalently, every abstract homomorphism from G to a compact Hausdorff group is trivial.

None of these conclusions requires fullness.

The first assertion upgrades the known uniqueness of the Polish topology to uniqueness among all separable SIN topologies, including incomplete ones. It also gives the following direct partial answer to the metric reformulation.

Corollary 2 (the SIN metric case). *Let M be a separable II_1 factor. If a compatible left-invariant metric d on $\text{PU}(M)$ induces a nontrivial separable SIN topology, then d is complete (and its topology is the strong operator topology). For a type III factor there is no such metric topology.*

The target paper's strong uncountable cofinality theorem converts the last part of Theorem 1 into a target-group result beyond SIN groups.

Corollary 3 (locally compact targets). *Let M be a factor with separable predual, of type II_1 or III. Every abstract homomorphism*

$$\text{PU}(M) \longrightarrow H$$

to a second-countable locally compact Hausdorff group H is trivial. No continuity of the homomorphism is assumed.

2 Proof intuition

The normal-generation estimates of Dowerk–Thom say uniformly that an element which stays a fixed positive distance from the scalars normally generates the whole projective unitary group in a bounded number of conjugates. In a SIN topology, sufficiently small neighbourhoods are invariant under conjugation. Such a neighbourhood therefore cannot contain an element bounded away from the scalars: otherwise a fixed power of the neighbourhood would be the whole group. Thus every SIN topology must be at least as fine as the canonical projective length topology.

For a II_1 factor this length topology is the strong topology. Invariant automatic continuity supplies the opposite inclusion for every separable SIN topology, giving equality. For a type III factor the canonical length is the projective operator norm, whose topology is nonseparable, so no separable SIN topology can refine it.

A precompact Hausdorff group topology is automatically SIN. If it refined one of the two canonical length topologies, that length topology would itself be precompact; explicit families of symmetries show that it is not. Hence the abstract groups have no nontrivial compact images. Finally, strong uncountable cofinality makes the image of any homomorphism into a group with a proper left-invariant metric bounded. In a locally compact second-countable target the image therefore has compact closure, and the compact-image result forces it to be trivial.

3 Proof

We isolate the general mechanism. An invariant length on a group G is a map $\ell : G \rightarrow [0, L]$ such that $\ell(1) = 0$, $\ell(g^{-1}) = \ell(g)$, $\ell(gh) \leq \ell(g) + \ell(h)$, and $\ell(hgh^{-1}) = \ell(g)$. We say that it has *uniform normal generation away from the identity* if, for every $\varepsilon > 0$, there is $N_\varepsilon \in \mathbb{N}$ such that

$$\ell(g) \geq \varepsilon \implies G = (g^G \cup g^{-G})^{N_\varepsilon}. \quad (1)$$

Lemma 4 (SIN refinement). *Suppose that ℓ is a Hausdorff invariant length satisfying (1). Every Hausdorff SIN group topology τ on G refines the topology defined by ℓ .*

Proof. Fix $0 < \varepsilon < L$ and put $N = N_\varepsilon$. Choose $a \neq 1$. Since τ is Hausdorff, there is a τ -neighbourhood O of 1 with $a \notin O$. Continuity of multiplication gives a symmetric neighbourhood U with $U^N \subseteq O$. The SIN property gives a symmetric conjugacy-invariant neighbourhood $V \subseteq U$.

If $v \in V$ and $\ell(v) \geq \varepsilon$, then $v^G \cup v^{-G} \subseteq V$, so (1) yields $G \subseteq V^N \subseteq O$, contradicting $a \notin O$. Hence $V \subseteq \{g : \ell(g) < \varepsilon\}$. Every ℓ -ball about 1 is therefore a τ -neighbourhood, which is the desired refinement. $\square$

For a II_1 factor with normalized trace τ_M , define on $G = \text{PU}(M)$

$$\ell_1([u]) = \inf_{\lambda \in \mathbb{T}} \tau_M(|1 - \lambda u|), \quad \ell_2([u]) = \inf_{\lambda \in \mathbb{T}} \|1 - \lambda u\|_2.$$

These define the same topology because

$$\ell_1([u]) \leq \ell_2([u]), \quad \ell_2([u])^2 \leq 2\ell_1([u]). \quad (2)$$

This is precisely the quotient strong operator topology. Dowerk–Thom’s normal-generation function for $\text{PU}(M)$ [2, Theorem 1.3] implies (1) for ℓ_1 : their bound is locally bounded when ℓ_1 is bounded below by a positive constant.

For a type III factor define

$$\ell_\infty([u]) = \inf_{\lambda \in \mathbb{T}} \|1 - \lambda u\|.$$

The optimal normal-generation estimate $2048/\ell_\infty$ from [3, Theorem 1.3] gives (1) for ℓ_∞ .

We also need two elementary noncompactness facts. A II_1 factor contains a trace-preserving diffuse abelian von Neumann algebra and hence Rademacher symmetries r_n . For $n \neq m$, the symmetry $r_n r_m$ takes the values 1 and -1 , each on a projection of trace $1/2$, and therefore

$$\ell_1([r_n^{-1} r_m]) = \inf_{\lambda \in \mathbb{T}} \frac{|1 - \lambda| + |1 + \lambda|}{2} = 1.$$

Thus the strong topology on $\text{PU}(M)$ is not precompact.

Every type III factor also contains a diffuse abelian von Neumann algebra. Choose a continuous increasing family of projections $(p_t)_{t \in (1/4, 3/4)}$ there and put $s_t = 1 - 2p_t$. If $s \neq t$, then $s_s s_t$ has both 1 and -1 in its spectrum, so

$$\ell_\infty([s_s^{-1} s_t]) = \inf_{\lambda \in \mathbb{T}} \max\{|1 - \lambda|, |1 + \lambda|\} = \sqrt{2}.$$

This uncountable separated family shows that the projective norm topology is nonseparable and, in particular, not precompact.

Proof of Theorem 1. The groups under consideration are algebraically simple [2, 3]; consequently every nontrivial group topology is Hausdorff.

In the II_1 case, Lemma 4 says that every Hausdorff SIN topology refines the strong topology. If the topology is also separable, invariant automatic continuity [2, Theorem 1.4], applied to the identity homomorphism from G with its strong topology to G with the new topology, gives the reverse refinement. The two topologies coincide.

In the type III case, Lemma 4 says that every Hausdorff SIN topology refines the projective norm topology. A countable dense set for the finer topology would then be dense for the norm topology, contradicting the uncountable $\sqrt{2}$ -separated family above.

Finally, every Hausdorff precompact group topology is SIN, because it embeds densely into its compact completion and compact groups are SIN. By Lemma 4 it refines the appropriate canonical length topology. Precompactness passes to a coarser group topology: for every canonical length ball B , which is open in the finer topology, finitely many translates of B cover G . This contradicts the separated families above. Hence no Hausdorff precompact topology exists.

If $\pi : G \rightarrow K$ is a nontrivial homomorphism to a compact Hausdorff group, algebraic simplicity makes π injective. Pulling the topology of K back along π gives a Hausdorff precompact group topology on G , which has just been ruled out. Thus π is trivial. $\square$

Proof of Corollary 2. Theorem 1 identifies the metric topology with the strong topology. The latter is induced by the complete bi-invariant projective 2-metric, so its left uniformity is complete. Any compatible left-invariant metric induces that same left uniformity and is therefore complete. The type III assertion is Theorem 1(2). $\square$

Proof of Corollary 3. By Struble’s theorem [4], H admits a proper compatible left-invariant metric d . Let G act on (H, d) by $g \cdot h = \pi(g)h$. This is an isometric action. Strong uncountable cofinality for G , proved in the source paper [1, Theorems 1.1 and 1.2] and inherited by projective quotients, implies that every orbit of every isometric action is bounded. Hence $\pi(G)$ is bounded in H . Properness of d makes its closure compact. The homomorphism into that compact closure is trivial by Theorem 1(3), so π is trivial. $\square$

4 Scope, upgrade attempts, and novelty check

The source problem remains open for arbitrary separable non-SIN topologies. The proof identifies two sharp restrictions on any counterexample: its completion must be non-locally-compact, and the topology cannot be SIN. Fullness does not bridge this gap: its spectral-gap formulation controls strong-operator central sequences, whereas an arbitrary neighbourhood need not have an invariant interior.

Six focused upgrade attempts were made. The successful ones were the SIN refinement lemma, the complete SIN classification, and the exclusion of all locally compact targets. Attempts to invariantize an arbitrary separable neighbourhood, use fullness to bound the support size in the known Steinhaus argument, obtain a countable-index-subgroup contradiction, and compactify an arbitrary bounded metric completion all fail at the same non-SIN obstruction. In particular, intersecting all conjugates of a neighbourhood need not leave interior, and strong uncountable cofinality gives boundedness rather than total boundedness.

Novelty searches through 17 August 2026 used the exact phrases from the source question, “unique separable SIN group topology”, “projective unitary group locally compact homomorphism”, the source arXiv id, and citations/keywords for the two Dowerk–Thom normal-generation papers. No explicit statement of Theorem 1 or Corollary 3 was found. The result is a short synthesis of known quantitative theorems, so novelty confidence is moderate even though proof confidence is high.

5 Verification and recommendation

The proof is noncomputational. The verifier should check: (i) that the cited normal-generation bounds are uniform on every set $\ell \geq \varepsilon$; (ii) the direction of both topology refinements in the Π_1 case; (iii) inheritance of SIN and precompactness from compact completions; and (iv) the use of

Struble properness plus abstract strong uncountable cofinality. Recommended disposition: human review as a likely-valid, substantial partial result.

References

- [1] P. A. Dowerk, *Strong uncountable cofinality for unitary groups of von Neumann algebras*, arXiv:1804.09515 (2018; revised 2019).
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- [4] R. A. Struble, *Metrics in locally compact groups*, Compositio Math. 28 (1974), 217–222.